

AI Coding Interview Assignment

Legal Management Module

Purpose of This Assignment

This assignment is designed to evaluate how effectively you can use modern AI coding agents and AI development tools to build a real-world software module.

We are not only evaluating whether you can write code manually. We are specifically evaluating your ability to:

- understand a complex product requirement,
- break it down into implementable parts,
- use AI coding tools productively,
- review and correct AI-generated code,
- make architectural decisions,
- test and validate the final result,
- explain your reasoning clearly.

You may use any AI coding agent or assistant, including but not limited to:

- Cursor, Claude Code, GitHub Copilot, ChatGPT, Windsurf, Codeium, Replit Agent, or other comparable AI coding tools.

The quality of your AI-assisted development workflow is part of the evaluation.

1. Product Context

We want to build an internal **Legal Management Module** for an organization's legal team. The module should act as a secure, organized, and auditable workspace for managing legal operations, including cases, contracts, notices, deadlines, tasks, documents, discussions, financial records, access control, activity logs, and offboarding.

The system should be treated as both a **system of record** for legal information and a **system of deadlines** for critical legal obligations. The module should support **Persian date handling** where relevant.

2. Assignment Goal

Build an MVP version of the **Legal Management Module**. You do not need a perfect enterprise product; your implementation should demonstrate strong product understanding, clean architecture, practical engineering choices, and effective use of AI coding tools.

3. Required Core Features

3.1 Authentication and Roles

Support at least: **Legal Admin**, **Legal Manager**, **Legal Counsel / Case Lead**, and **Viewer**. Demonstrate clear permission boundaries.

3.2 Legal Case Management

Include: title, reference code, type, status, priority, owner, involved parties, description, dates, documents, and timeline.

3.3 Contract Management

Include: title, type, counterparty, status, owner, effective/expiration/renewal dates, documents, and key terms.

3.4 Legal Notice Management

Track notice intake: sender, received date, response deadline, status, and related entities.

3.5 Deadline and Reminder Management

Critical feature. Track deadlines for cases, contracts, and notices. Provide views for: upcoming, overdue, today's, and assigned-to-me deadlines.

3.6 Tasks and Collaboration

Simple task management (title, description, assignee, due date, status) linked to legal matters.

3.7 Documents

Store metadata (name, type, entity link, uploaded by, date, description) with enforced access control based on record permissions.

3.8 Activity Log / Audit Trail

Log critical actions (created, updated, reassigned, status changed) with actor and timestamp.

3.9 Dashboard

A centralized view (respecting permissions) summarizing cases, contracts, notices, and deadlines.

3.10 Reassignment and Offboarding

Support transferring ownership of work. Bonus for a bulk offboarding flow.

4. Suggested Screens / API Areas

Build a UI or an API-first project (e.g., /auth/login, /dashboard, /cases, /contracts, /notices, /deadlines, /tasks, /documents, /activity-logs).

5. Technical Expectations

Choose your preferred stack (e.g., Next.js, Django, FastAPI, etc.). Include clear setup instructions, seed data, readable structure, and testing.

6. AI Tool Usage Requirement

You must use AI tools for requirement breakdown, schema design, boilerplate, testing, and refactoring. You are responsible for the final output; do not submit code you do not understand.

7. Required AI Usage Documentation

Include an AI_USAGE.md file covering:

1. **Tools Used**
2. **Development Workflow**
3. **Important Prompts** (5–10)
4. **AI Mistakes or Limitations** (At least 2)
5. **What You Personally Decided** (Your professional judgment)

8. Required Deliverables

1. Source code (GitHub/GitLab link or ZIP)
2. README.md
3. AI_USAGE.md
4. Database migration/schema
5. Seed data
6. Setup and running instructions
7. Tests

9. Timebox

1 to 2 days. Prioritize clarity, architecture, and AI-assisted workflow over feature breadth.

10. Evaluation Rubric

We assess:

- **Product Understanding** (Confidentiality, auditability, legal context)
- **Technical Design** (Architecture, data modeling, validation)
- **AI Usage** (Verification, prompting strategy, correction)
- **Code Quality** (Readability, security, modularity)
- **Testing & Verification** (Correctness of core logic)
- **Prioritization** (Smart tradeoffs)

11. Candidate Presentation

Be prepared for a 15–20 minute presentation covering:

- Problem understanding
- Module decomposition

- Tooling and workflow
- Verification methods
- Future improvements